

Dr. Sumit Kumar Roy

Research publication portfolio

Affiliation	Assistant Professor, Department of Physics, St. Xavier’s College Ranchi, Jharkhand, India
Research areas	Ceramics; nanotechnology; ferroelectrics
Scholar metrics	124 citations; h-index 8; i10-index 5
Coverage	20 works represented; publication years 2017–2026
Profile	Google Scholar record

Year-wise publication matrix

Rows are shaded by complete year blocks. Position is the researcher’s place in the abbreviated author list (position/total). A dash indicates that the corresponding value was not identified.

Year	Author pos.	Authors (abbrev.)	Paper title	Received / preprint*	Published	Journal / publication
2026	4/5	R. Kumar; R. K. Singh; K. Mamta; S. K. Roy; S. K. Pandey	Ultra-low thermal conductivity and dielectric stability in novel high-entropy perovskite oxides for advanced thermal and electronic applications	–	2026	Journal of Materials Science: Materials in Electronics 37(3), 264
2025	4/5	S. Rajwar; R. K. Singh; S. K. Singh; S. K. Roy; K. Prasad	Tuning the β -phase in BNT/PVDF composite films: structural and impedance study	–	2025	Journal of Polymer Research
2025	4/4	M. Kumar; R. K. Singh; S. K. Mukherjee; S. K. Roy	First-principles investigation of structure, electronic and optical properties of newly synthesized double perovskite $\text{Nd}_2\text{FeTiO}_6$ for optoelectronic applications	–	2025	Processing and Application of Ceramics
2025	3/3	R. K. Singh; M. Kumar; S. K. Roy	Exploration of eco-friendly multifunctional material: investigating the dielectric, ferroelectric, and optical properties of $\text{Bi}_{0.5}(\text{Na}_{0.78}\text{K}_{0.22})_{0.5}\text{TiO}_3$	–	2025	Journal of Materials Science: Materials in Electronics
2024	4/6	K. Sweta; A. Singh; S. Kumari; S. K. Roy; S. K. Pandey; K. Prasad	Electrical properties of $\text{Ba}_{0.06}(\text{Na}_{0.5}\text{Bi}_{0.5})_{0.94}\text{TiO}_3\text{--Ba}(\text{Fe}_{0.5}\text{Ta}_{0.5})\text{O}_3$ ceramics: Impedance spectroscopy analysis	–	2024	AIP Conference Proceedings
2024	3/5	R. K. Singh; M. Kumar; S. K. Roy; S. K. Pandey; D. Kumar	Electrical analysis of lead free $\text{Bi}_{0.5}(\text{Na}_{0.78}\text{K}_{0.22})_{0.5}\text{TiO}_3$ ceramic	–	2024	AIP Conference Proceedings
2024	3/4	M. Kumar; R. K. Singh; S. K. Roy; D. Kumar	Impedance spectroscopy of newly synthesized double perovskite ceramic: $\text{Nd}_2\text{FeTiO}_6$	–	2024	AIP Conference Proceedings
2023	3/4	S. Lakra; S. K. Mukherjee; S. K. Roy; M. Kumar	Dielectric and Mechanical Properties of PZT Incorporated PVDF Nanocomposites Prepared by Solution Casting Method	–	2023	Materials Science Forum
2023	3/3	R. K. Singh; M. Kumar; S. K. Roy	Structure, Microstructure, and Dielectric Properties of $\text{Bi}_{0.5}(\text{Na}_{0.78}\text{K}_{0.22})_{0.5}\text{TiO}_3$ Lead-Free Ceramic	–	2023	Materials Science Forum

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2023	3/3	S. Rajwar; S. K. Pandey; S. K. Roy	Impedance Spectroscopy Analysis of (1- x)Ba _{0.06} (Na _{1/2} Bi _{1/2}) _{0.94} TiO ₃ - x NaTaO ₃ Ceramics	–	2023	Materials Science Forum
2023	2/4	M. Kumar; S. K. Roy; R. K. Singh; K. Prasad	Double perovskite Nd ₂ FeTiO ₆ ceramics: Structural and electrical properties	–	2023	Processing and Application of Ceramics
2023	4/4	S. Pandey; S. Chaudhuri; R. K. Singh; S. K. Roy	Effect of Milling on Dielectric Properties of PZT	–	2023	Materials Science Forum
2023	4/5	P. K. Mahato; S. Choudhuri; C. Kumar; S. Roy; P. Patra	Evaluation of crystal size present in graphene oxide quantum dots using optical and Raman spectroscopy	–	2023	Materials Today: Proceedings
2020	1/3	S. K. Roy; S. Chaudhuri; K. Prasad	Impedance spectroscopy analysis of (1- x)Ba _{0.06} (Na _{1/2} Bi _{1/2}) _{0.94} TiO ₃ - x NaNbO ₃ ceramics	–	2020	AIP Conference Proceedings
2020	2/2	S. Chaudhuri; S. K. Roy	Milling maps of nanocrystalline lead titanate ceramics synthesised by high energy ball milling	–	2020	AIP Conference Proceedings
2020	1/2	S. K. Roy; K. Prasad	Evaluation of Structural and Electrical Properties of Ball-milled NaNbO ₃ Ceramic	–	2020	Nano Progress
2019	1/4	S. K. Roy; S. N. Singh; S. K. Mukherjee; K. Prasad	Structure and dielectric studies of (1- x)Ba _{0.06} (Na _{0.5} Bi _{0.5}) _{0.94} TiO ₃ - x Ba(Fe _{0.5} Nb _{0.5})O ₃ lead-free ceramics	–	2019	Processing and Application of Ceramics
2018	1/2	S. K. Roy; K. Prasad	Hydrothermal nanotechnology: putting the last first	–	2018	Exploring the Realms of Nature for Nanosynthesis (book chapter)
2018	2/3	U. K. Mahto; S. K. Roy; K. Prasad	High energy milled Ba _{0.06} Na _{0.47} Bi _{0.47} TiO ₃ ceramic: Structural and electrical properties	–	2018	IEEE Transactions on Dielectrics and Electrical Insulation
2017	1/4	S. K. Roy; S. N. Singh; S. K. Mukherjee; K. Prasad	Ba _{0.06} (Na _{1/2} Bi _{1/2}) _{0.94} TiO ₃ -Ba(Fe _{1/2} Ta _{1/2})O ₃ : giant permittivity lead-free ceramics	–	2017	Journal of Materials Science: Materials in Electronics

Reading notes. Alternating pale-blue and white bands apply to complete profile-year groups. Author position is the researcher's place in the abbreviated author list (position/total). No arXiv preprints were identified for this researcher, and received or submitted dates were not recorded on the available publisher records, so the corresponding column is dashed throughout. Author order follows the displayed Google Scholar record, with the two author strings compressed in the source shown in expanded form; therefore a profile year can differ from the later journal publication year.

Data snapshot. Public Google Scholar profile and linked publisher records, checked 26 August 2026. Scholar metrics are dynamic and may change after this snapshot.